

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)	(i)		all four carbonates (1) acids react with metal carbonates to form carbon dioxide gas (1)	2			2		2
		(ii)		(hydrochloric acid would form) <u>insoluble</u> compound with Pb^{2+} (1) should use nitric acid / ethanoic acid (1)			2	2		2
		(iii)		award (2) for all four correct award (1) for any two correct $\text{Mg}^{2+}(\text{aq})$ white (precipitate) $\text{Fe}^{2+}(\text{aq})$ (dark) green (precipitate) $\text{Cr}^{3+}(\text{aq})$ (grey) green (precipitate) $\text{Pb}^{2+}(\text{aq})$ white (precipitate)	2			2		2
		(iv)		add excess sodium hydroxide (1) award (1) for either of following - magnesium hydroxide white precipitate remains but lead hydroxide precipitate dissolves (giving a colourless solution) - iron(II) hydroxide green precipitate remains but chromium hydroxide precipitate dissolves (giving a dark green solution) lead and chromium are amphoteric (iron and magnesium are not) (1)			3	3		3